Day 72 – Network Troubleshooting & Fault Isolation

Overview

Day 72 shifted the focus from **building networks** to **diagnosing and repairing networks**.

Throughout Days 65–71, the primary objective was learning how networking technologies function and how to configure them. By Day 72, the focus changed to answering a different question:

> What happens when the network stops working?

Rather than introducing new networking technologies, this lab introduced a structured troubleshooting methodology used by network engineers, systems engineers, and cloud engineers to identify and resolve failures.

The goal was not simply to fix problems, but to learn how to systematically isolate root causes and avoid random guessing.

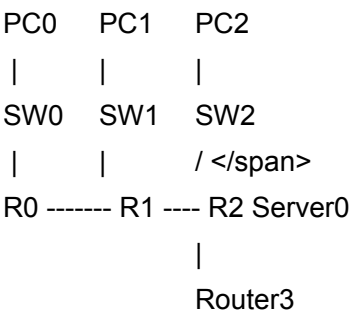
Objectives

By the end of this lab, I was able to:

- Understand structured troubleshooting methodology.
- Diagnose physical and logical network failures.
- Troubleshoot interface failures.
- Troubleshoot routing protocol failures.
- Troubleshoot ACL-related connectivity issues.
- Use Cisco verification commands effectively.
- Differentiate between multiple failure types that produce similar symptoms.
- Apply a repeatable troubleshooting workflow.

Existing Enterprise Topology

The enterprise network created throughout Days 65–71 was used.



Services already operating:

✅ RIP

✅ DHCP

✅ NAT

✅ ACLs

✅ Server Infrastructure

✅ ISP Simulation

Troubleshooting Philosophy

One of the most important lessons of Day 72 was learning that troubleshooting should follow a process rather than random configuration changes.

Poor troubleshooting:

Change settings.
Hope something works.

Professional troubleshooting:

Observe symptoms

↓

Gather evidence

↓

Identify root cause

↓

Implement fix

↓

Verify solution

The Layered Troubleshooting Model

Network issues should be investigated from the bottom up.

Layer 1 – Physical

Questions:

Is the cable connected?
Is the interface up?
Is the device online?

Primary command:

show ip interface brief

Layer 2 – Switching

Questions:

Correct VLAN?
Correct switch port?
Correct broadcast domain?

Primary commands:

show vlan brief
show mac address-table

Layer 3 – Routing

Questions:

Does the router know where to send traffic?
Are routes present?

Primary commands:

show ip route
show ip protocols

Layer 4 and Above – Services

Questions:

DHCP issue?
ACL issue?
DNS issue?
NAT issue?

Primary commands:

show access-lists
show ip dhcp binding

Lab 1 – Interface Failure

Scenario

Router0’s router-to-router interface was intentionally disabled.

Command:

interface g0/1
shutdown

What Was Broken?

Router0 lost communication with Router1.

Visual:

Before:

PC0
↓
R0
↓
R1
↓
R2
↓
PC2

After:

PC0
↓
R0
❌
R1
R2
PC2

The path beyond Router0 no longer existed.

Investigation

Command:

show ip interface brief

Finding:

GigabitEthernet0/1
administratively down

Diagnosis

The keyword:

administratively down

indicated that the interface had been manually disabled through configuration.

Resolution

Command:

interface g0/1
no shutdown

Verification:

G0/1
up/up

Connectivity path restored.

Lab 2 – RIP Failure

Scenario

Router1 was instructed to stop participating in RIP on the router-to-router networks.

Command:

router rip
no network 10.0.0.0

What Was Broken?

Router1 stopped advertising and receiving RIP updates on the transit links connecting the routers.

Unlike Lab 1:

Interfaces remained up/up

making the failure harder to detect.

Investigation

Command:

show ip protocols

Finding:

Before:

Routing for Networks:
10.0.0.0
192.168.20.0

After:

Routing for Networks:
192.168.20.0

The:

10.0.0.0

network was missing.

Diagnosis

Router1 was no longer exchanging routing information on the transit network.

This was a routing failure, not an interface failure.

Resolution

Command:

router rip
network 10.0.0.0

Verification:

Routing for Networks:
10.0.0.0
192.168.20.0

RIP functionality restored.

Lab 3 – ACL Failure

Scenario

A restrictive ACL was intentionally deployed.

Configuration:

access-list 99 deny any
interface g0/0

ip access-group 99 out

What Was Broken?

Traffic destined for LAN30 was blocked.

Unlike Lab 2:

Routes still existed.

Unlike Lab 1:

Interfaces still existed.

Yet communication failed.

Investigation

Command:

show access-lists

Finding:

Standard IP access list 99
10 deny any

Hit counter:

4 match(es)

Diagnosis

The ACL was actively preventing traffic from reaching LAN30.

The hit count confirmed packets were being denied.

Resolution

Command:

interface g0/0
no ip access-group 99 out

Verification:

Traffic resumed successfully.

Lab 4 – Troubleshooting Toolkit Review

The final lab consolidated the troubleshooting process and reviewed the most important commands learned throughout Phase 4.

Interface Verification

show ip interface brief

Used to determine:

Interface Status
up/up
down/down
administratively down

Routing Verification

show ip route

Used to determine:

Known Networks
Connected Routes
RIP Routes

RIP Verification

show ip protocols

Used to determine:

Routing Protocol Status
Participating Networks
Routing Neighbors

ACL Verification

show access-lists

Used to determine:

Permit Rules
Deny Rules
ACL Hit Counters

DHCP Verification

show ip dhcp binding

Used to determine:

Active DHCP Leases

NAT Verification

show ip nat translations

Used to determine:

Address Translation Activity

VLAN Verification

show vlan brief

Used to determine:

Existing VLANs
Port Membership

Key Lessons Learned

Interface Problems

Symptoms:

No connectivity

Troubleshooting:

show ip interface brief

Routing Problems

Symptoms:

Remote networks unreachable

Troubleshooting:

show ip route
show ip protocols

ACL Problems

Symptoms:

Traffic blocked despite healthy interfaces and routes

Troubleshooting:

show access-lists

Most Important Lesson

Different failures can produce similar symptoms.

For example:

Failed Ping

may result from:

- Interface failure
- Routing failure
- ACL restriction
- DHCP issue
- NAT issue

A successful engineer identifies the actual cause before attempting a fix.

Troubleshooting Workflow

The most valuable takeaway from Day 72 was the troubleshooting process itself:

1. Observe symptoms
2. Gather information
3. Isolate the problem
4. Identify root cause
5. Apply fix

6. Verify solution

This methodology is more valuable than any individual command because it applies to networking, systems administration, cloud engineering, and cybersecurity.

What I Learned

- Troubleshooting is a structured process.
- Failed connectivity can have many causes.
- Interface problems are diagnosed differently than routing problems.
- Routing problems are diagnosed differently than ACL problems.
- Verification commands are essential troubleshooting tools.
- Evidence should guide troubleshooting decisions.
- Random configuration changes create more problems than solutions.
- Network engineers add value by identifying root causes, not simply by configuring devices.

Reflection

Day 72 was one of the most practical labs in Phase 4 because it focused on diagnosing problems rather than building services. By intentionally breaking interfaces, routing protocols, and ACLs, I learned how to isolate failures using evidence-based troubleshooting methods. The lab reinforced the importance of verification commands and demonstrated that similar symptoms can originate from very different root causes. This experience helped transition my thinking from configuration-focused networking to operational network support and fault isolation.

Day 72 Status

✅ Interface Failure Troubleshooting Completed

✅ RIP Failure Troubleshooting Completed

✅ ACL Failure Troubleshooting Completed

✅ Troubleshooting Toolkit Reviewed

✅ Verification Commands Mastered

✅ Root Cause Analysis Practiced

✅ Structured Troubleshooting Methodology Learned

✅ Ready for **Day 73 – Network Monitoring & Management** 🚀